
Let M be a matroid and $e \in M$. The polynomial $\hat{Y}_M(x)$ is defined as

$$\hat{Y}_M(x) = \sum_{F \in \mathcal{L}(M)} \left(x^{\mathrm{crk}(F)} \mu[F, E] \right) \cdot \hat{Q}_{M^F}(x).$$

For $F \in \mathcal{L}(M)$, we have,

$$\hat{Y}_{M^F}(x) = \sum_{G \leq F} \left(x^{\mathrm{rk}(F) - \mathrm{rk}(G)} \mu[G, F] \right) \cdot \hat{Q}_{M^G}(x)$$

and

$$\hat{Y}_{M/G}(x) = \sum_{F \geq G} \left(x^{\mathrm{rk}(E) - \mathrm{rk}(F)} \mu[F, E] \right) \cdot \hat{Q}_{M^F_G}(x).$$

We want to find a formula for $Y_M(x) := (-1)^{\mathrm{rk}(M)} \hat{Y}_M(x)$ in terms of $Y_{M \setminus e}(x)$.

Suggestion 1

Let $\mu_F := \sum_{G \leq F} x^{\mathrm{rk}(G)} \mu[G, F] \cdot G$. Then, writing μ_M in Kazhdan-Lusztig basis, we have

$$\begin{aligned}
\mu_M &= \sum_{F \in \mathcal{L}(M)} \left(\left(x^{\mathrm{rk}(F)} \mu[F, E] \right) \cdot \sum_{G \leq F} x^{\mathrm{rk}(F) - \mathrm{rk}(G)} \hat{Q}_{M^F_G}(x^{-2}) \cdot \zeta^G \right) \\
&= \sum_{F \in \mathcal{L}(M)} \sum_{G \leq F} x^{2\mathrm{rk}(F) - \mathrm{rk}(G)} \mu[F, E] \cdot \hat{Q}_{M^F_G}(x^{-2}) \cdot \zeta^G \\
&= \sum_{G \in \mathcal{L}(M)} \sum_{F \geq G} x^{2\mathrm{rk}(F) - \mathrm{rk}(G)} \mu[F, E] \cdot \hat{Q}_{M^F_G}(x^{-2}) \cdot \zeta^G \\
&= \sum_{G \in \mathcal{L}(M)} x^{-\mathrm{rk}(G)} \left(\sum_{F \geq G} x^{2\mathrm{rk}(F)} \mu[F, E] \cdot \hat{Q}_{M^F_G}(x^{-2}) \right) \cdot \zeta^G
\end{aligned}$$

The exponent of x in the above expression should be $-2\mathrm{crk}(F) = 2\mathrm{rk}(F) - 2\mathrm{rk}(M)$ for us to get the expression for μ_M in terms of $\hat{Y}_{M/G}(x)$.

Redefining μ_F as $\sum_{G \leq F} x^{\mathrm{rk}(G) - 2\mathrm{rk}(F)} \mu[G, F] \cdot G$, we have

$$\begin{aligned}
\mu_M &= \sum_{G \in \mathcal{L}(M)} x^{-\mathrm{rk}(G)} \left(\sum_{F \geq G} x^{2\mathrm{rk}(F) - 2\mathrm{rk}(E)} \mu[F, E] \cdot \hat{Q}_{M^F_G}(x^{-2}) \right) \cdot \zeta^G \\
&= \sum_{G \in \mathcal{L}(M)} x^{-\mathrm{rk}(G)} \hat{Y}_{M/G}(x^{-2}) \cdot \zeta^G
\end{aligned}$$

Computing $\Delta(\mu_M)$

We defined $\mu_M = \sum_{F \in \mathcal{L}(M)} x^{\mathrm{rk}(F)-2\mathrm{rk}(E)} \mu[F, E] \cdot F$. Then,

$$\begin{aligned}\Delta(\mu_M) &= \sum_{F \ni e \wedge F \setminus e \in \mathcal{L}(M)} x^{\mathrm{rk}(F)-2\mathrm{rk}(E)-1} \mu[F, E] \cdot (F \setminus e) + \sum_{\text{other terms}} x^{\mathrm{rk}(F)-2\mathrm{rk}(E)} \mu[F, E] \cdot (F \setminus e) \\ &= \sum_{F \in S} x^{\mathrm{rk}(F)-2\mathrm{rk}(E)} \mu[F \cup e, E] \cdot F + \mu_{M \setminus e}\end{aligned}$$

The desired expression for $\Delta(\mu_M)$ is $\mu_{M \setminus e}$ which we could not obtain.